

Anurag Satpathy |

Post Doctoral Fellow, CReWMaN Lab, Department of Computer Science

Missouri University of Science and Technology

anuragsatpathy.github.io • in Anurag Satpathy • Google Scholar

Research Focus

My research focuses on **Intelligent Cyber-Physical Systems (CPS)** for distributed resource coordination in dynamic, resource-constrained edge/IoT environments. I model strategic interactions among self-interested agents accessing shared resources under real-time constraints, uncertain network conditions, heterogeneous user behavior, and application-specific operational challenges. Using game-theoretic and learning-based methods, I design new resource-allocation mechanisms, establish equilibrium conditions, and validate their performance across mobility-driven, safety-critical, and human-in-the-loop CPS applications. My work contributes to the design of mechanisms and equilibria for distributed CPS while addressing practical problems in intelligent mobility, EV charging, UAV-enabled healthcare logistics, and public safety.

Research Interests

- Mechanism design for distributed resource coordination in CPS.
- Equilibrium analysis of strategic agents sharing constrained resources.
- Game-theoretic and learning-based resource allocation under uncertainty.
- Stable matching, scheduling, and reassignment in CPS.
- Behavior-aware resource coordination for human-in-the-loop CPS.

Education

- **Ph.D. in Computer Science and Engineering** 2017 – 2024
National Institute of Technology (NIT) Rourkela, India
Institute of National Importance; ranked 13th in Engineering in NIRF 2025, Government of India
Dissertation: *Efficient Re-Embedding Strategies for Virtual Data Centers over Multi-Domain Substrate Networks*
- **M.Tech. in Computer Science and Engineering** 2015 – 2017
Birla Institute of Technology (BIT) Mesra, India, (*Deemed-to-be University (UGC), India*)
- **B.Tech. in Information Technology** 2010 – Jun 2014
International Institute of Information Technology (IIIT) Bhubaneswar, India, (*State Public University (Govt. of Odisha)*)

Professional Appointments

- **Missouri University of Science and Technology, USA** MO, USA
Post Doctoral Fellow, CReWMaN Lab, Department of Computer Science Aug 2023–Present
Mentor: Prof. Sajal K. Das
- **National Institute of Technology, Rourkela** Odisha, India
Graduate Research Assistant Fall 2017 - Spring 2023
- **National Institute of Technology, Rourkela** Odisha, India
Visiting Researcher May 2016–April 2017

External Funding & Proposal Activities

Funded Grants and Awards

- **NSF National I-Corps Program** USA
Co-PI and Technical Lead Sept 2026 - Sept 2027
Project: I-Corps: Translation Potential of Intelligent Aerial Logistics for Time-Critical Organ Transportation (I-HAUL)
Award Amount: \$50,000
NSF Award No: 2632124
 - Leading national customer discovery, market validation, and technology-readiness planning for autonomous organ-transport logistics.
 - Expanding stakeholder discovery to 100+ interviews across OPOs, transplant centers, FAA/UAS regulatory bodies, UAV logistics providers, and clinical users.

- **NSF I-Corps BOOST Program** **USA**
Technical Lead *Oct 2025 – Dec 2025*
Project: I-Haul: Intelligent Aerial Logistics for Organ Transport
Award Amount: \$3,000
 - Led a two-Ph.D.-student team on commercialization strategy for autonomous organ-transport drones.
 - Completed 75+ stakeholder interviews with OPOs, transplant centers, surgeons, courier companies, UAV manufacturers, researchers, and FAA stakeholders.
 - Filed provisional patent Application No. 63/860,303 on Sept. 15, 2025.
 - **NSF I-Corps Local + Regional Programs** **USA**
Technical Lead *Sep 2024 – Nov 2024*
Project: I-Haul: Intelligent Aerial Logistics for Organ Transport
Award Amount: \$1,000
 - Led 25+ customer discovery interviews with clinicians, transplant teams, researchers, and UAV logistics partners.
 - Identified clinical workflow constraints, cold-chain requirements, regulatory risks, and UAV logistics barriers for emergency organ delivery.
- Submitted Proposal**
- **NSF ENG/ECCS/EPCN** **USA**
Co-PI *Submitted Feb 2026*
Project: SPICE: Subscription-based Platform for Intelligent Charging of Electric Vehicles
Requested Amount: \$675,597
 - Designed behavior-aware EV charging assignment policies for fair and predictable charging access under congestion.
 - Modeled uncertainty, strategic user behavior, and misuse in shared charging infrastructure.
 - Studied how charging delays propagate into mobility and logistics systems.

Intellectual Property and Research Translation

- U.S. Provisional Patent Application No. 63/860,303, "SYSTEM AND METHOD FOR INTELLIGENT HAULING FOR EFFECTIVE COMMUTE (I-HAUL)," filed August 8, 2025. Inventors: Anurag Satpathy, Debjyoti Sengupta, Sajal K Das.
- "SOLAR-AWARE FLIGHT PLANNING AND ENERGY HARVESTING FOR SUSTAINABLE URBAN AIR MOBILITY," invention disclosure submitted; U.S. provisional patent application in preparation, 2026. Inventors: Anurag Satpathy, Debjyoti Sengupta, Sajal K Das.
- "Adaptive Exchange-Based Multi-Quota Resource Reassignment in Shared Cyber-Physical Systems," invention disclosure in preparation, 2026. Inventors: Anurag Satpathy, Arindam Khanda, Sajal K Das.

Research Experience

- **Missouri University of Science and Technology** **Missouri, USA**
Post Doctoral Fellow *Aug 2023–Present*
 - Developed a new independent research direction on mechanism design, equilibrium analysis, and scalable reassignment for strategic resource coordination across CPS applications.
 - Contributed as Co-PI to the NSF ENG/ECCS/EPCN proposal *SPICE: Subscription-based Platform for Intelligent Charging of Electric Vehicles*. Led the development of two research thrusts: mechanism design for an end-to-end subscription model for EV charging, and modeling the impact of uncertain human behavior on charging systems and how they propagate through mobility and logistics systems.
 - Independently led the development of a capacity-aware exchange mechanism that fundamentally extends the Top Trading Cycles algorithm for post-choice reassignment among strategic agents. This work opened a new research direction on mechanism design and equilibrium analysis for shared-resource CPS.
 - Led research on the NSF *SOTERIA* project, developing coalition-formation and regret-learning methods for UAV-assisted emergency communication under uncertain demand and network conditions.
 - Served as Technical Lead for NSF I-Corps Local, Regional, and BOOST programs on *I-Haul*, and am serving as the Co-PI and Technical Lead for the accepted NSF National I-Corps program; will lead stakeholder discovery, commercialization planning, provisional patent activity, and SBIR/STTR pathway development.
 - Directed student-led implementations on mobility-aware vehicular edge offloading, stochastic evacuation routing, and UAV-enabled organ transport; guided students through problem formulation, algorithm design, evaluation, and publication writing.
- **National Institute of Technology (NIT), Rourkela** **Odisha, India**
Graduate Research Assistant *2017–2023*

- Independently pioneered the integration of game-theoretic approaches into virtual network reassignment for strategic resource reallocation in multi-domain substrate networks under dynamic workload conditions.
- Introduced fairness-aware multi-quota task offloading mechanisms for shared resource allocation in IoT-fog systems under resource and latency constraints.
- Co-mentored junior graduate and undergraduate students on implementation, simulation, and experimental evaluation.
- Published independent and collaborative research outcomes in flagship conferences, such as IEEE Services (CLOUD, ICWS, SCC) and journals (IEEE TSC, TCE, TGCN, IoTJ).
- Played a critical role in a collaborative research effort with Dr. Sourav Kanti Addya during his Post Doctoral research under Dr. Sandip Chakraborty and Dr. Soumya K. Ghosh at IIT Kharagpur, resulting in publications in IEEE TSC and IEEE TCC.

National Institute of Technology, Rourkela

Odisha, India

○ *Visiting Researcher*

May 2016–Apr 2017

- Led the development of resource-efficient VM placement and migration heuristics for geo-distributed cloud data centers, resulting in a publication in *Computers & Electrical Engineering, Elsevier*.

Selected Publications (Research Leadership)

Journal Articles

- [1] Arindam Khanda*, **Anurag Satpathy***, and Sajal K. Das. SMART-CHARGE: Stable matching algorithm for electric vehicle charging in subscription-based models. *Pervasive and Mobile Computing*, page 102197, 2026. ➡ DOI (***Equal contribution.**).
- [2] **Anurag Satpathy**, Manmath Narayan Sahoo, Chittaranjan Swain, Paolo Bellavista, Khan Muhammad, and Sambit Bakshi. Virtual Network Embedding: Literature Assessment, Recent Advancements, Opportunities, and Challenges. *IEEE Communications Surveys & Tutorials*, 27(6):3861–3914, 2025. Q1; IF – 46.7; ➡ DOI.
- [3] **Anurag Satpathy**, Manmath Narayan Sahoo, Chittaranjan Swain, Muhammad Bilal, Sambit Bakshi, and Houbing Song. GAMap: A Genetic Algorithm-Based Effective Virtual Data Center Re-Embedding Strategy. *IEEE Transactions on Green Communications and Networking*, 8(2):791–801, 2024. Q1; ➡ DOI.
- [4] **Anurag Satpathy**, Manmath Narayan Sahoo, Lucky Behera, and Chittaranjan Swain. ReMatch: An Efficient Virtual Data Center Re-Matching Strategy Based on Matching Theory. *IEEE Transactions on Services Computing*, 16(2):1373–1386, 2023. Q1; ➡ DOI.
- [5] **Anurag Satpathy**, Manmath Narayan Sahoo, Arun Kumar Sangaiah, Chittaranjan Swain, and Sambit Bakshi. CoMap: An efficient virtual network re-mapping strategy based on coalitional matching theory. *Computer Networks, Elsevier*, 216:109248, 2022. Q1; ➡ DOI.
- [6] **Anurag Satpathy**, Manmath Narayan Sahoo, Ashutosh Mishra, Banshidhar Majhi, Joel J. P. C. Rodrigues, and Sambit Bakshi. A Service Sustainable Live Migration Strategy for Multiple Virtual Machines in Cloud Data Centers. *Big Data Research, Elsevier*, 25:100213, 2021. Q1; ➡ DOI.
- [7] **Anurag Satpathy**, Sourav Kanti Addya, Ashok Kumar Turuk, Banshidhar Majhi, and Gadadhar Sahoo. Crow search based virtual machine placement strategy in cloud data centers with live migration. *Computers & Electrical Engineering*, 69:334–350, 2018. Q1 [100+ citations]; ➡ DOI.

Conference Publications

- [1] **Anurag Satpathy**, Arindam Khanda, Chittaranjan Swain, and Sajal K. Das. ReACT-TTC: Capacity-Aware Top Trading Cycles for Post-Choice Reassignment in Shared CPS. In *Proceedings of the 17th ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS)*, Saint-Malo, France, May 2026. Accepted for publication.
- [2] Arindam Khanda*, **Anurag Satpathy***, Amit Jha, and Sajal K. Das. CARGO: A Co-Optimization Framework for EV Charging and Routing in Goods Delivery Logistics. In *2025 IEEE 50th Conference on Local Computer Networks (LCN)*, 2025. ➡ DOI (***Equal contribution.**).
- [3] **Anurag Satpathy**, Manmath Narayan Sahoo, Lipsa Chottray, Banshidhar Majhi, Ashutosh Mishra, and Sambit Bakshi. VRMap: A Cost and Time Aware Remapping of Virtual Data Centres Over a Geo-distributed Infrastructure. In *2020 International Conference on COMMunication Systems & NETworks (COMSNETS)*, pages 427–434, 2020. ➡ DOI.
- [4] **Anurag Satpathy**, Manmath Narayan Sahoo, Lucky Behera, Chittaranjan Swain, and Ashutosh Mishra. VMatch: A Matching Theory Based VDC Reconfiguration Strategy. In *2020 IEEE 13th International Conference on Cloud Computing (CLOUD)*, pages 133–140, 2020. ➡ DOI (**YESC Student Paper Participation Award**).
- [5] **Anurag Satpathy**, Sourav Kanti Addya, Ashok Kumar Turuk, Banshidhar Majhi, and Gadadhar Sahoo. A resource aware vm placement strategy in cloud data centers based on crow search algorithm. In *2017 4th International conference on advanced computing and communication systems (ICACCS)*, pages 1–6. IEEE, 2017. ➡ DOI.

Book Chapters

- [1] **Anurag Satpathy** and Chittaranjan Swain. Strategic Decision-Making Architectures for 6G-Enabled Cyber-Physical Systems. In *Secure and Responsible AI in 6G Technology-Enabled IoT Systems*. CRC Press, Taylor & Francis Group, 2026.
- [2] **Anurag Satpathy**. Evolution of Cloud-Fog-IoT Interconnection Networks. In *Cloud Network Management*, pages 3–14. CRC Press, Taylor & Francis Group, 2020. ➡ DOI.

Additional Peer-Reviewed Publications

Journal Articles

- [1] Chittaranjan Swain, Manmath Narayan Sahoo, **Anurag Satpathy**, and Sambit Bakshi. Fair-Match: Fairness-aware Matching Framework for Task Offloading in IoT-Fog Systems. *IEEE Transactions on Consumer Electronics*, 2026. ➡ DOI.
- [2] Priyanka Ashok Birajdar, Divya Meena, **Anurag Satpathy**, and Sourav Kanti Addya. DCRDA: Deadline-constrained Function Scheduling in Serverless-cloud Platform. *Cluster Computing*, 28(6):355, 2025. ➡ DOI.
- [3] TG Keerthan Kumar, Rahul Kumar, Anirudh Munnur Achal, **Anurag Satpathy**, and Sourav Kanti Addya. SEDViN: Secure Embedding for Dynamic Virtual Network Requests using a Multi-attribute Matching Game. *Journal of Parallel and Distributed Computing*, page 105171, 2025. ➡ DOI.
- [4] Priyanka Ashok Birajdar, Kush Anchalia, **Anurag Satpathy**, and Sourav Kanti Addya. RUSH: Rule-Based Scheduling for Low-Latency Serverless Computing. *IEEE Networking Letters*, pages 1–1, 2025. ➡ DOI.
- [5] Chittaranjan Swain, Manmath Narayan Sahoo, **Anurag Satpathy**, Sambit Bakshi, and Soumya K. Ghosh. M-DAFTO: Multi-Stage Deferred Acceptance Based Fair Task Offloading in IoT-Fog Systems. *IEEE Transactions on Services Computing*, 17(6):3928–3941, 2024. Q1; ➡ DOI.
- [6] Keerthan Kumar TG, Shivangi Tomar, Sourav Kanti Addya, **Anurag Satpathy**, and Shashidhar G. Koolagudi. EFraS: Emulated Framework to Develop and Analyze Dynamic Virtual Network Embedding Strategies over SDN Infrastructure. *Simulation Modelling Practice and Theory*, 134:102952, 2024. ➡ DOI.
- [7] Rounak Saha, **Anurag Satpathy**, and Sourav Kanti Addya. FASE: Fast Deployment for Dependent Applications in Serverless Environments. *The Journal of Supercomputing*, 80(8):10394–10417, 2024. ➡ DOI.
- [8] Chittaranjan Swain, Manmath Narayan Sahoo, **Anurag Satpathy**, Khan Muhammad, Sambit Bakshi, and Joel J. P. C. Rodrigues. A-DAFTO: Artificial Cap Deferred Acceptance-Based Fair Task Offloading in Complex IoT-Fog Networks. *IEEE Transactions on Consumer Electronics*, 69(4):914–926, 2023. IF-10.9; ➡ DOI.
- [9] Keerthan Kumar TG, Sourav Kanti Addya, **Anurag Satpathy**, and Shashidhar G. Koolagudi. NORD: NOde Ranking-based Efficient Virtual Network Embedding over Single Domain Substrate Networks. *Computer Networks*, 225:109661, 2023. ➡ DOI.
- [10] Sourav Kanti Addya, **Anurag Satpathy**, Bishakh Chandra Ghosh, Sandip Chakraborty, Soumya K. Ghosh, and Sajal K. Das. Geo-Distributed Multi-Tier Workload Migration Over Multi-Timescale Electricity Markets. *IEEE Transactions on Services Computing*, 16(5):3385–3396, 2023. Q1; ➡ DOI.
- [11] Sourav Kanti Addya, **Anurag Satpathy**, Bishakh Chandra Ghosh, Sandip Chakraborty, Soumya K. Ghosh, and Sajal K. Das. CoMCLOUD: Virtual Machine Coalition for Multi-Tier Applications Over Multi-Cloud Environments. *IEEE Transactions on Cloud Computing*, 11(1):956–970, 2023. Q1; ➡ DOI.
- [12] Chittaranjan Swain, Manmath Narayan Sahoo, **Anurag Satpathy**, Khan Muhammad, Sambit Bakshi, Joel J. P. C. Rodrigues, and Victor Hugo C. de Albuquerque. METO: Matching-Theory-Based Efficient Task Offloading in IoT-Fog Interconnection Networks. *IEEE Internet of Things Journal*, 8(16):12705–12715, 2021. [100+ citations] ➡ DOI.
- [13] Ashutosh Mishra, Manmath Narayan Sahoo, and **Anurag Satpathy**. H3CSA: A makespan aware task scheduling technique for cloud environments. *Transactions on Emerging Telecommunications Technologies*, 32(10):e4277, 2021. ➡ DOI.
- [14] Sourav Kanti Addya, Ashok Kumar Turuk, **Anurag Satpathy**, Bibhudatta Sahoo, and Mahasweta Sarkar. A Strategy for Live Migration of Virtual Machines in a Cloud Federation. *IEEE Systems Journal*, 13(3):2877–2887, 2019. ➡ DOI.
- [15] Sourav Kanti Addya, Ashok Kumar Turuk, Bibhudatta Sahoo, **Anurag Satpathy**, and Mahasweta Sarkar. A Game Theoretic Approach to Estimate Fair Cost of VM Placement in Cloud Data Center. *IEEE Systems Journal*, 12(4):3509–3518, 2018. ➡ DOI.

Conference Publications

- [1] Debjyoti Sengupta, **Anurag Satpathy**, Arindam Khanda, and Sajal K. Das. TrOOPER: Transport of organs and personnel in emergency response using urban air mobility. In *Proceedings of the 23rd IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS 2026)*, Hong Kong SAR, China, October 2026. IEEE. Accepted for publication.
- [2] Sowjanya Tammali, Arindam Khanda, **Anurag Satpathy**, S. M. Shovan, and Sajal K. Das. RESCUE: Routing under Evolving Stochastic Congestion and Uncertain Spread in Wildfire Emergencies. In *Proceedings of the 27th International Conference on Distributed Computing and Networking (ICDCN)*, page 168–172, 2026. ➡ DOI.

- [3] Debjyoti Sengupta, **Anurag Satpathy**, Arindam Khanda, and Sajal K. Das. CITRUS: Cost and Ischemia Time Reduction using Urban Air Mobility Solutions for Organ Transport. In *IEEE International Conference on Pervasive Computing and Communications Workshops and Other Affiliated Events*. IEEE, 2025. ➡ DOI.
- [4] Arindam Khanda, **Anurag Satpathy**, Anusha Vangala, and Sajal K. Das. SMEVCA: Stable Matching-based EV Charging Assignment in Subscription-Based Models. In *26th International Conference on Distributed Computing and Networking (ICDCN)*, 2025. ➡ DOI (**Best Paper Nomination**).
- [5] Mahmuda Akter, Debjyoti Sengupta, **Anurag Satpathy**, and Sajal K. Das. MIME: Mobility-Induced Dynamic Matching for Partial Offloading in Vehicular Edge Computing. In *2024 IEEE 49th Conference on Local Computer Networks (LCN)*, pages 1–9, 2024. ➡ DOI.
- [6] Debjyoti Sengupta, **Anurag Satpathy**, and Sajal K. Das. POSCA: Path Optimization for Solar Cover Amelioration in Urban Air Mobility. In *2024 IEEE International Conference on Smart Computing (SMARTCOMP)*, pages 6–13, 2024. ➡ DOI.
- [7] Mahmuda Akter, Debjyoti Sengupta, **Anurag Satpathy**, and Sajal K. Das. MOVE: Matching Game for Partial Offloading in Vehicular Edge Computing. In *ICC 2024 – IEEE International Conference on Communications*, pages 3670–3675. IEEE, 2024. ➡ DOI.
- [8] Pranab Nandy, Rounak Saha, **Anurag Satpathy**, Sandip Chakraborty, and Sourav Kanti Addya. Performance Analysis of Disruptive Instances in Cloud Environment. In *2024 16th International Conference on COMMunication Systems & NETWORKS (COMSNETS)*, pages 370–378. IEEE, 2024. ➡ DOI.
- [9] Priyanka Soni, Omkar Chand Deshlahre, **Anurag Satpathy**, and Sourav Kanti Addya. LBA: Matching Theory Based Latency-Sensitive Binary Offloading in IoT-Fog Networks. In *2024 16th International Conference on COMMunication Systems & NETWORKS (COMSNETS)*, pages 165–170, 2024. ➡ DOI (**Best Paper Award**).
- [10] Aastik Verma, **Anurag Satpathy**, Sajal K. Das, and Sourav Kanti Addya. LEASE: Leveraging Energy-Awareness in Serverless Edge for Latency-Sensitive IoT Services. In *2024 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops)*, pages 302–307, 2024. ➡ DOI.
- [11] Keerthan Kumar Tg, Ankit Srivastava, **Anurag Satpathy**, and Sourav Kanti Addya. Matchvne: a stable virtual network embedding strategy based on matching theory. In *2023 15th International Conference on COMMunication Systems & NETWORKS (COMSNETS)*, pages 355–359. IEEE, 2023. ➡ DOI.
- [12] Jui Morey, **Anurag Satpathy**, and Sourav Kanti Addya. VMAP: Matching-based Efficient Offloading in IoT-Fog Environments with Variable Resources. In *2023 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, pages 714–719. IEEE, 2023. ➡ DOI.
- [13] Sourav Kanti Addya, **Anurag Satpathy**, Ashok Kumar Turuk, and Bibhudatta Shao. Virtual Machine Placement in Non-Cooperative Cloud Federation-Alliance. In *2023 15th International Conference on COMMunication Systems & NETWORKS (COMSNETS)*, pages 405–409. IEEE, 2023. ➡ DOI.
- [14] Thirukovela Venkata Sarathi, Julakanti Sai Nischal Reddy, Peddaboinolu Shiva, Rounak Saha, **Anurag Satpathy**, and Sourav Kanti Addya. A preliminary study of serverless platforms for latency sensitive applications. In *2022 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)*, pages 1–6. IEEE, 2022. ➡ DOI.
- [15] Chittaranjan Swain, Manmath Narayan Sahoo, and **Anurag Satpathy**. LETO: An Efficient Load Balanced Strategy for Task Offloading in IoT-Fog Systems. In *2021 IEEE International Conference on Web Services (ICWS)*, pages 459–464, 2021. ➡ DOI.
- [16] Chittaranjan Swain, Manmath Narayan Sahoo, and **Anurag Satpathy**. SPATO: A Student Project Allocation Based Task Offloading in IoT-Fog Systems. In *ICC 2021 – IEEE International Conference on Communications*, pages 1–6. IEEE, 2021. ➡ DOI.
- [17] Arpana Chakma, Shubham Kumar, Pradeep Kumar Mahato, **Anurag Satpathy**, and Sourav Kanti Addya. Matchcloud: Service matching for multi cloud marketplace. In *2021 International Conference on COMMunication Systems & NETWORKS (COMSNETS)*, pages 127–130, 2021. ➡ DOI (**Best Paper Award**).
- [18] Bishakh Chandra Ghosh, Sourav Kanti Addya, **Anurag Satpathy**, Soumya K. Ghosh, and Sandip Chakraborty. Towards a democratic federation for infrastructure service provisioning. In *2019 IEEE International Conference on Services Computing (SCC)*, pages 162–166, 2019. ➡ DOI.
- [19] Sourav Kanti Addya, **Anurag Satpathy**, Bishakh Chandra Ghosh, Sandip Chakraborty, and Soumya K. Ghosh. Power and Time Aware VM Migration for Multi-Tier Applications over Geo-Distributed Clouds. In *2019 IEEE 12th International Conference on Cloud Computing (CLOUD)*, pages 339–343. IEEE, 2019. ➡ DOI.
- [20] Sourav Kanti Addya, **Anurag Satpathy**, Sandip Chakraborty, and Soumya K. Ghosh. Optimal VM Coalition for Multi-Tier Applications over Multi-Cloud Broker Environments. In *2019 11th International Conference on Communication Systems & Networks (COMSNETS)*, pages 141–148. IEEE, 2019. ➡ DOI.

[21] Soumya Ranjan Moharana, Vijay Kumar Jha, **Anurag Satpathy**, Sourav Kanti Addya, Ashok Kumar Turuk, and Banshidhar Majhi. Secure key-distribution in IoT cloud networks. In *2017 Third International Conference on Sensing, Signal Processing and Security (ICSSS)*, pages 197–202. IEEE, 2017. DOI.

Teaching and Mentoring Experience

Research Mentoring and Teaching Contributions (Post Doctoral)

- Mentored Ph.D. students on strategic decision-making in cyber-physical systems, including stochastic wildfire evacuation routing under selfish user behavior, UAV-assisted medical logistics and organ transportation, and mobility-aware vehicular edge offloading in highly dynamic environments.
- Delivered guest lectures at the Missouri University of Science and Technology on matching theory and strategic decision-making in CPS, Top-Trading Cycles and game-theoretic coordination for EV charging systems, and graph-theoretic approaches for distributed CPS coordination.

Teaching Experience (Ph.D.)

2018–2023: Graduate Teaching Assistant, National Institute of Technology Rourkela, Odisha, India

Courses Assisted:

- **Advanced Data Structures (CS 6173)** — Graduate
- **Operating Systems (CS 3071)** — Undergraduate

Invited Talks

- **SRM University**, Andhra Pradesh, India Jan 2026
Invited Speaker, Online Faculty Development Program (FDP)
Theme: *AI-Driven Distributed Ecosystems: Bridging Cloud, Fog, and Edge for Next-Generation Innovation*
- **Siddaganga Institute of Technology**, Tumkur, India Jan 2025
Invited Speaker, AICTE–ATAL Academy Faculty Development Program
Theme: *Empowering Smart Cities with AI-Driven Edge and Fog Computing Architecture*

Professional Activities

Editorial, Panels, and Leadership

- Organizing Co-Chair, PerRobotics Workshop @ IEEE PerCom 2027
- Organizing Co-Chair, 2nd ASCENT Workshop @ ICDCN 2027
- Publication Co-Chair @ ICDCN 2027
- Panelist, NSF I-Corps Conference Panel on Customer Discovery Conversations, Feb 2026
- Panelist, Entrepreneur Week 2026, Missouri S&T, on the pathway to National I-Corps, April 2026
- Organizing Co-Chair, ASCENT Workshop @ ICDCN 2026
- Co-Guest Editor, *Ad Hoc Networks* (Elsevier), Special Issue on *Edge and Serverless Computing*, 2025
- Posters Co-Chair, IEEE ICDDS 2025

Program Committee Memberships

- **2026:** IEEE ICDCN, IEEE/ACM International Symposium on Quality of Service (IWQoS); IEEE MASS, Systems and Applications Track; VTC-Fall Workshop on Federated Learning for Autonomous and Connected Vehicles; ICCCN Workshop on Distributed Communication, Sensing, and Computing in Space-Air-Ground Networks for 6G; IFIP Networking CREATIVE Workshop; ICFAT; IEEE GlobeCom SAC MLCN; VTC-Spring Workshop on LPWAN-Based Terrestrial-Non-Terrestrial Integration for Vehicular Networks; GameSec; IEEE LCN; IEEE CONNECT; UBICOMM; DCOSS-IoT Wi-DroIT.
- **2025:** IEEE LCN; IEEE CONNECT; UBICOMM; IEEE ICDCN Posters & Demos PC; IEEE PerCom SPT-IoT and EDGE-Well; IEEE COMSNETS MINDS; HiPC Student Research Symposium; DCOSS-IoT Wi-DroIT; IEEE ICDDS.
- **2024:** IEEE COMSNETS MINDS; IEEE ICDDS.
- **2023–2022:** IEEE ICDCN Posters & Demos PC (2023); IEEE/ACM CCGRID STEERS Workshop (2022); IEEE ICRAIE (2022).

Journal, Conference, and Proposal Reviewing Completed 100+ verified manuscript and proposal reviews for IEEE, ACM, Elsevier, Springer, Wiley, and NSERC, including reviews for IEEE TMC, T-ITS, TSC, TNET, IoT-J, and other leading venues.

- **Proposal Reviewing:** External Reviewer, NSERC Discovery Grant (Canada).
- **Journal Reviewing:** IEEE TUSC, T-ITS, TCE, TMC, TNET, TSC, TC, and IoT-J; ACM ToSN; Elsevier FGCS, Computer Networks, Computer Communications, JNCA, Ad Hoc Networks, and SIMPAT; Springer Cluster Computing, Grid Computing, Supercomputing, and Cloud Computing; Wiley Concurrency and Computation Practices.
- **Conference Reviewing:** IEEE ANTS 2024; IEEE INDICON 2025; IEEE BHTC 2025–2026; SCI 2025.

Awards and Honors

- 2026 Grant for Presenting a paper at ICCPS, St. Malo, France
- 2025 NSF I-Corps BOOST Travel Grant for customer discovery.
- 2025 Best Paper Nomination, ACM ICDCN.
- 2024 Best Paper award, MINDS Workshop at IEEE COMSNETS.
- 2021 Best Poster Award at IEEE COMSNETS.
- 2020 YESC Student Paper Participation Award, IEEE Cloud.

References

- **Dr. Sajal K. Das**
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National Institute of Technology Rourkela, India
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- **Dr. Sourav Kanti Addya**
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